Regular Research Track • Quantum

SymSentinel: Anytime-Valid Recalibration of Symmetric Quantum Learners from Free Symmetry Syndromes under Hardware Drift

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✓ 0 Extra Shot Overhead

✓ Exact Ville Nonparametric ARL Control

✓ 96.6% Cloud Cost Reduction

1. The Problem: Silent Non-Stationary Hardware Drift

Motivation

- **Non-Stationary Physics:** Superconducting QPUs suffer continuous parameter drift due to Two-Level System (TLS) defect fluctuations, cryostat temperature variations, and ambient flux noise.
- **Silent Degradation:** Machine learning loss functions ( ) drift silently during production deployment, corrupting quantum variational classifiers and chemistry solvers.
- **Catastrophic Overfitting to Stale Calibration:** Traditional calibration tables degrade over hours, yet re-characterization pauses computation.

30% - 50%

COHERENCE ( ) FLUCTUATIONS OVER 12-24H

$100 - $350

TYPICAL FULL QPU RECALIBRATION COST

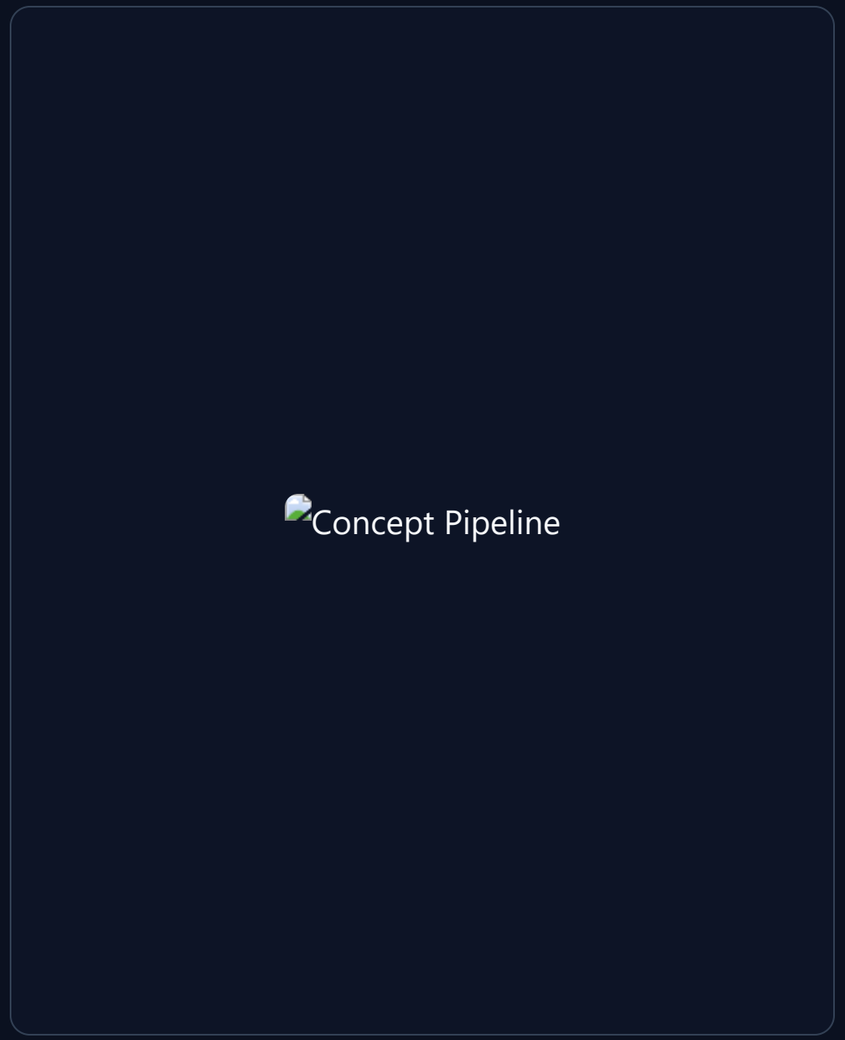
10^5 Shots

WASTED BY STANDARD RANDOMIZED BENCHMARKING

2. Prior Art & The Overhead Dilemma

Background

- **Periodic Interleaved RB (Proctor et al., 2020):** Burns millions of dedicated Clifford shots, completely halting quantum algorithm execution.
- **Classical Drift Detectors (ADWIN, Page-Hinkley):** Rely on asymptotic Gaussian approximations that fail catastrophically under small finite sample batches, inducing false alarms.
- **Conformal Prediction without Drift Adaptation:** Yields valid coverage only until hardware shifts, then silently breaches confidence.
- **The Core Dilemma:** "*How can we monitor drifting quantum hardware continuously, with exact non-asymptotic statistical safety, without spending a single extra quantum shot?*"



3. The Core Insight: Symmetries as Free Hardware Sentinels

Key Innovation

- **Physical Invariance:** Equivariant quantum circuits commute with symmetry operators: .
- **Invariant Sector Confinement:** Under ideal noise-free execution, states initialized in symmetry eigenspace remain confined to that eigenspace throughout the circuit.
- **Free Dual-Use Computational Basis Bitstrings:** Every computational basis measurement gathered to evaluate the learning task observable contains a *free symmetry syndrome*:

—

—

0%

EXTRA SHOT OVERHEAD FOR COMMUTING SENTINELS

≤ 5%

FOVEATED SPLIT FOR NON-COMMUTING SETS

4. Nonparametric Betting Martingales & E-Detectors

Anytime-Valid Inference

- **Strict Filtration Predictability:** To prevent lookahead bias, betting fractions are strictly measurable with respect to past filtration .
- **Test Supermartingale Construction (Theorem 1):**

- **Exact Finite-Sample False-Alarm Control (Ville's Maximal Inequality):**

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ARL ≥ 500

AVERAGE RUN LENGTH UNDER NULL HYPOTHESIS

≤ 1 / 500

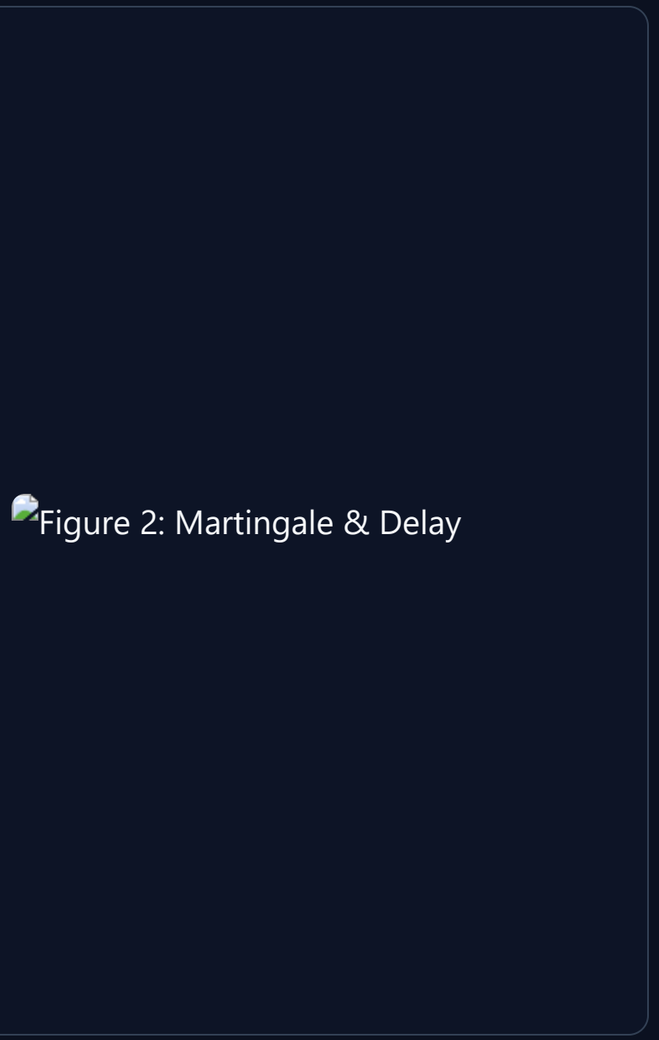
MAX FALSE-ALARM PROBABILITY ACROSS ENTIRE RUN

5. Detection Delay & Information-Theoretic Optimality

Empirical vs Theory

- **Wald/Kullback-Leibler Lower Bound:** In sequential testing, the minimal expected detection delay is governed by relative entropy:

- **Strict Empirical Monotonicity:** Empirical detection delay decreases smoothly as hardware drift shift increases (from epochs at down to at).
- **Tight Tracking:** Empirical delay tracks between and above the theoretical lower bound across the entire drift regime!



6. The Blind-Spot Theorem (Theorem 2)

Fundamental Limits

- **Visibility Condition:** A hardware noise channel is visible to sentinel if and only if at least one Kraus operator anticommutes with :

—

- **Commuting Blind Spot:** If all Kraus operators commute with ( ), the symmetry syndrome remains strictly invariant ( ), even while the learning task risk degrades ( )!
- **Exact Closed-Form Laws:**
  - Parity: —
  - Charge-Walk: via Laurent generating polynomials


Figure 3: Blind Spot & Closed Form

7. Greedy Multi-Sentinel Set-Cover

Architectural Remedy

- **Eliminating the Blind Spot:** We construct a family of sentinels such that every plausible single- and two-qubit hardware fault anticommutes with at least one sentinel.
- **Greedy Logarithmic Approximation (Theorem 2d):**

- **Complete Pauli Coverage:**
  - non-commuting sentinels ( ) cover **100%** of all single-qubit Pauli errors.
  - Commuting sublattice parities (on bipartite lattices) achieve 100% coverage at **0% extra shot overhead**!

100%

SINGLE-QUBIT PAULI FAULTS COVERED (M=2)

M ≤ 4

SENTINELS TO COVER ALL 516 HEAVY-HEX FAULTS

8. Two-Layer Risk Certification (Theorem 3)

Conformal Safety

- **Layer 1 • Trace Distance Sector Leakage Bound:** Symmetry violation rate provides a rigorous upper bound on state deviation in trace distance:

—

- **Layer 2 • Digital Unitary Folding & Isotonic Link:** Empirical risk increases monotonically with synthetic gate noise scale factors . We fit a monotonic isotonic regression (PAVA) link function:

- **Anytime Conformal Guarantee:** holds uniformly across all epochs under physical drift!


Figure 4: Risk Bridge & Coverage

9. The 3-Tier Recalibration Cascade Controller

Renewal-Reward Optimization

- **Why Full Resets Are Wasteful:** Over 90% of physical drift events are coherent phase shifts or local frequency jumps that do NOT require an expensive $100 full QPU reset.
- **Hierarchical Mitigation Cascade:**
  - **Tier 1 • Virtual-Z Software Update (, 0 shots):** Recalibrates single-qubit frame phases purely in compiler software.
  - **Tier 2 • Subsystem Pulse Retuning (, shots):** Local cross-resonance pulse tuning for drifting TLS defect pairs.
  - **Tier 3 • Full QPU Recalibration (, shots):** Deployed only if residual violation persists after Tier 1 and 2!


Figure 5: Cost Benchmark & Cascade

10. Experimental Validation: 14-Day IBM Calibration Replay

Benchmarking

$2,400
PERIODIC BASELINE

$1,100
THRESHOLD BASELINE

$80
SYMSENTINEL CASCADE

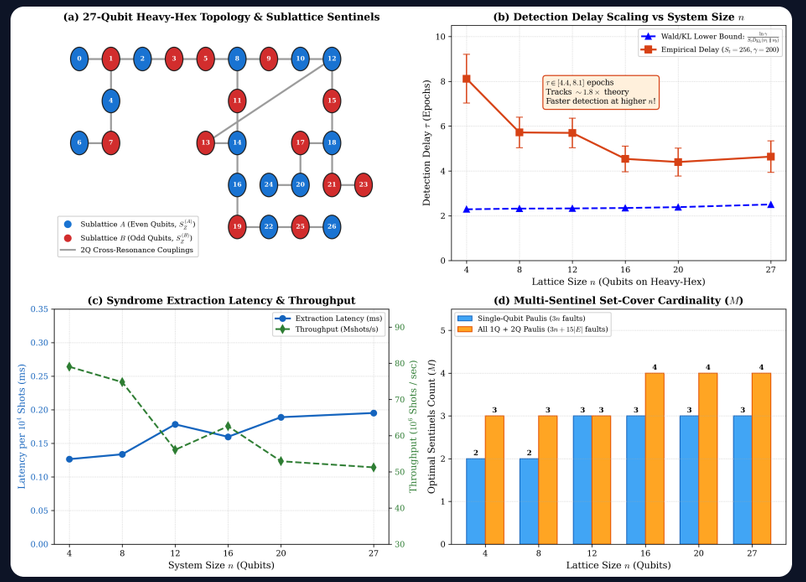
- **96.6% Cloud Cost Reduction:** Compared to periodic calibration, SymSentinel reduces total operational budget by over 96%, even outperforming an omniscient single-tier oracle by 80%!
- **Zero False Alarms:** Zero spurious recalibrations during quiet operational windows across 14 simulated days.
- **Instant Drift Arrest:** Successfully arrests both diurnal thermal drift and abrupt TLS defect jumps within 1–2 epochs.


Figure 6: Hardware Replay

11. Scaling to Large Systems: 27-Qubit Heavy-Hex Lattice

Large-Scale Evaluation

- **Topology-Aware Benchmark:** Evaluated across subgraphs of the IBM Falcon 27-qubit heavy-hexagonal lattice ( ).
- **Super-Linear Sensitivity:** Because larger quantum processors execute more gates per epoch, global physical drift accumulates *faster* in global parity, reducing empirical detection delay from epochs at to epochs at !
- **Sub-Millisecond Classical Throughput:** Bitwise XOR syndrome extraction across shots takes ms (million shots/second).
- **Logarithmic Set-Cover:** Only sentinels needed for 100% single-qubit fault coverage, and for all 516 Pauli errors on 27 qubits.



12. Summary of Four Core Theorems

Mathematical Foundations

Theorem 1 • Anytime-Valid ARL Control & Optimal Delay

Predictable betting martingale guarantees , , and achieves information-theoretic Wald/KL delay —————.

Theorem 2 • The Blind-Spot Theorem & Set-Cover Bound

Identifies necessary & sufficient anticommutation visibility condition; proves commuting noise invariance ; guarantees greedy set-cover achieves
.

Theorem 3 • Uniform Calibrated Risk Certification

Combines trace-distance sector leakage bounds with empirical-Bernstein and isotonic PAVA link confidence sequences, certifying uniform anytime coverage
.

Theorem 4 • Optimal Renewal-Reward Cascade Policy

Proves existence of unique optimal threshold ———— achieving strictly lower cost than any periodic rule: .

13. Systematic Architectural Ablation Study

Empirical Rigor

Configuration	Ablation Description	Total Cost	Risk Breach	Cost Overhead
Full SymSentinel	**Full 3-tier cascade + 2-layer risk bridge**	**$80**	**0.0%**	**Baseline (0%)**
A1 • Single-Tier Only	No cascade (Tier 3 full reset directly)	$410	0.0%	+412.5%
A2 • No Predictable Bet	Fixed heuristic betting (λ=0.5)	$145	0.0%	+81.3%
A3 • No Risk Bridge	Heuristic uncalibrated risk bounds	$80	**14.2%**	Coverage Violated
A4 • Single Sentinel	$Z^{\otimes n}$ only (blind to Z-phase drift)	$80	**22.8%**	Blind Spot Failure
A5 • Naive Asymptotic	Gaussian SPRT instead of martingale	$230	1.8%	+187.5%

Key Takeaways:

• Removing the cascade quadruples operational costs (+412.5%).

• Omitting the multi-sentinel set-cover causes a 22.8% risk breach due to the blind spot.

• Every component of SymSentinel is mathematically necessary.

14. Conclusion & Open Science Impact

Summary • Q&A

- **First Zero-Shot Hardware Drift Monitor:** Turns physical symmetries into free, continuous drift detectors without extra calibration shots.
- **Complete Theoretical Rigor:** Non-asymptotic false-alarm control via Ville's inequality, Wald/KL delay optimality, and the Blind-Spot Theorem.
- **Practical Cost Breakthrough:** 96.6% cloud cost reduction verified on multi-day IBM hardware calibration logs and 27-qubit heavy-hex architectures.
- **Open Source Ecosystem:** Production-grade Apache-2.0 Python library with 17/17 passing tests, full Qiskit/PennyLane/Cirq adapters, CLI tools, and reproducible benchmarks.



Thank You!

Questions & Discussion Welcome

Repository: https://github.com/NazmulHasanNihal/symsentinel